Ontologies, Multiagent Digital Twin and Intelligent Telemetry for Internet Infrastructure (DT-II)

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IFSPe (PE), ITA (SP), UFABC (SP), UFRGS(RS), and UNICAP (PE)

cgi.br

Knowledge Base (Ontology)

The Knowledge Base

Knowledge Base (Ontology)

Description Logic (OWL)

The Knowledge Base

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Knowledge Base (Ontology)

Description Logic (OWL)

TBox

Terminological Box

ABox

Assertion Box

RBox

Role Box

The Knowledge Base

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Description Logic (OWL)

TBox

Terminological Box

Concept Definitions

- Person ⊑ Mammal[(1)]
- Car ⊑ Vehicle[(2)]
- Driver ≡ Person who drives a vehicle

ABox

Assertion Box

Facts about individuals

- Mary is a Person
- Fiat is a Car
- Mary drivex Fiat

RBox

Role Box

Relationship Rules

- drives ⊑ interactsWith[(3)]
- hasFather ⊑ hasParent
- hasBrother is symmetric[(4)]
- married is functional[(5)]

(1) Every person is a mammal
(2) Every car is a vehicle
(3) Driving is a type of interaction
(4) If A is B's brother, then B is A's brother
(5) Each person is married to at most one other person

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Types of Ontologies

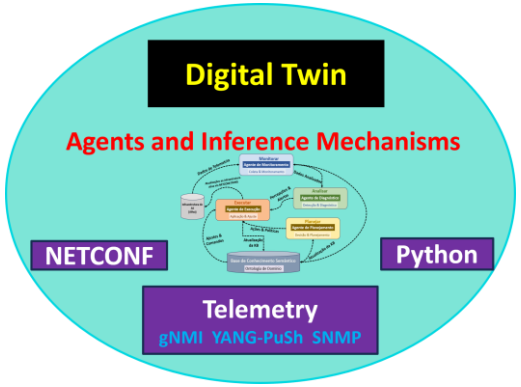
- **Upper Ontology**: highly general and domain-independent concepts, including object, event, process, quality, time, and space. Examples include BFO, DOLCE, UFO, SUMO, and GFO.
- **Domain Ontology**: concepts of a specific area.

UFO in the Development of Domain Ontologies

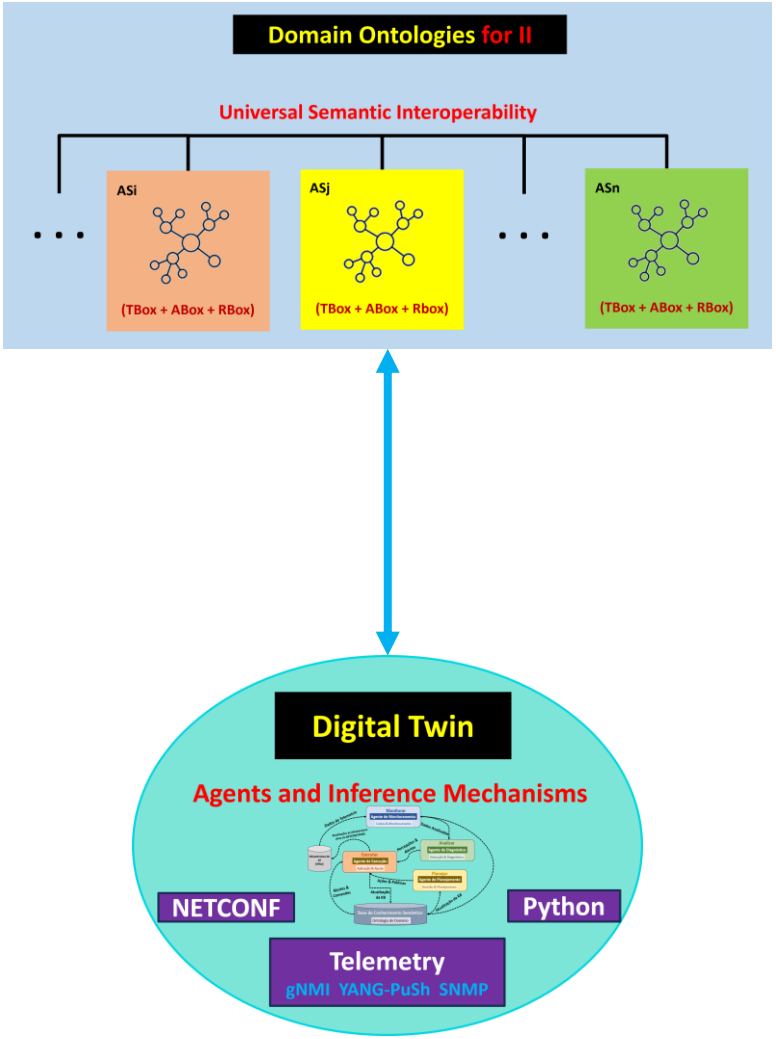
Advantages	In the Generic Domain	In the Information Infrastructure Domain
Distinguishing what something is from the role it plays	A person remains a person throughout their life, but is a student only for a period, while enrolled. It may seem obvious, yet information systems often confuse this distinction. UFO enforces the need to make it explicit.	A network is an Autonomous System, but simultaneously acts as a provider to one neighbour and as a client to another. Without this separation, the model fails as soon as the same network assumes two roles.
Being able to state when something continues to be "the same"	A football team replaces all its players over the years and yet remains the same team. UFO requires one to explicitly state what ensures this continuity.	An AS replaces routers, cables, and addresses, yet remains the same AS. What sustains it is the official registry that created it, not the number nor the equipment.
Treating agreements and relations as "real things"	A marriage is not merely a line connecting two people. It has a start date, rules, and may end. UFO treats such relations as entities in their own right, called *relators*.	A traffic exchange agreement between two networks has a beginning, conditions, and an end. Modelled in this way, the system can answer *since when have these networks been partners?*
Providing a common foundation that enables interoperability with other models	If two cities use the same socket standard, any appliance works in both. Ontologies built upon UFO share the same conceptual "socket standard", thus fitting together with less effort.	An AS ontology based on UFO can be linked to existing ontologies of contracts, services, or risks, rather than starting entirely from scratch.

UFO entails higher initial costs, yet prevents costly errors along the way. It proves worthwhile when the domain is complex, involves relations and agreements, and when the model must endure and integrate with others, as is the case

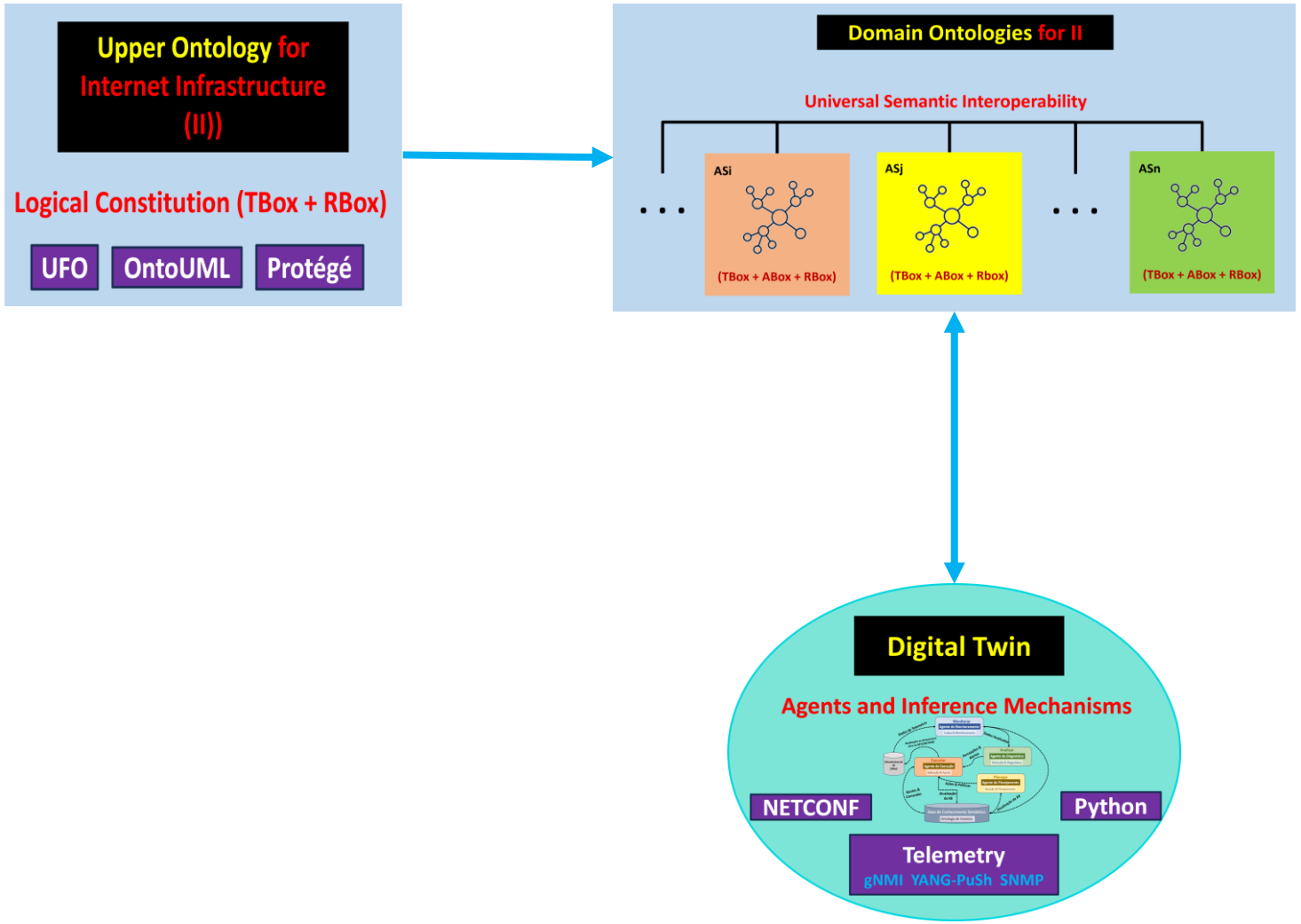
The Project


Digital Twin
Agents and Inference Mechanisms
NETCONF
Python
Telemetry
gNMI YANG-PuSh SNMP

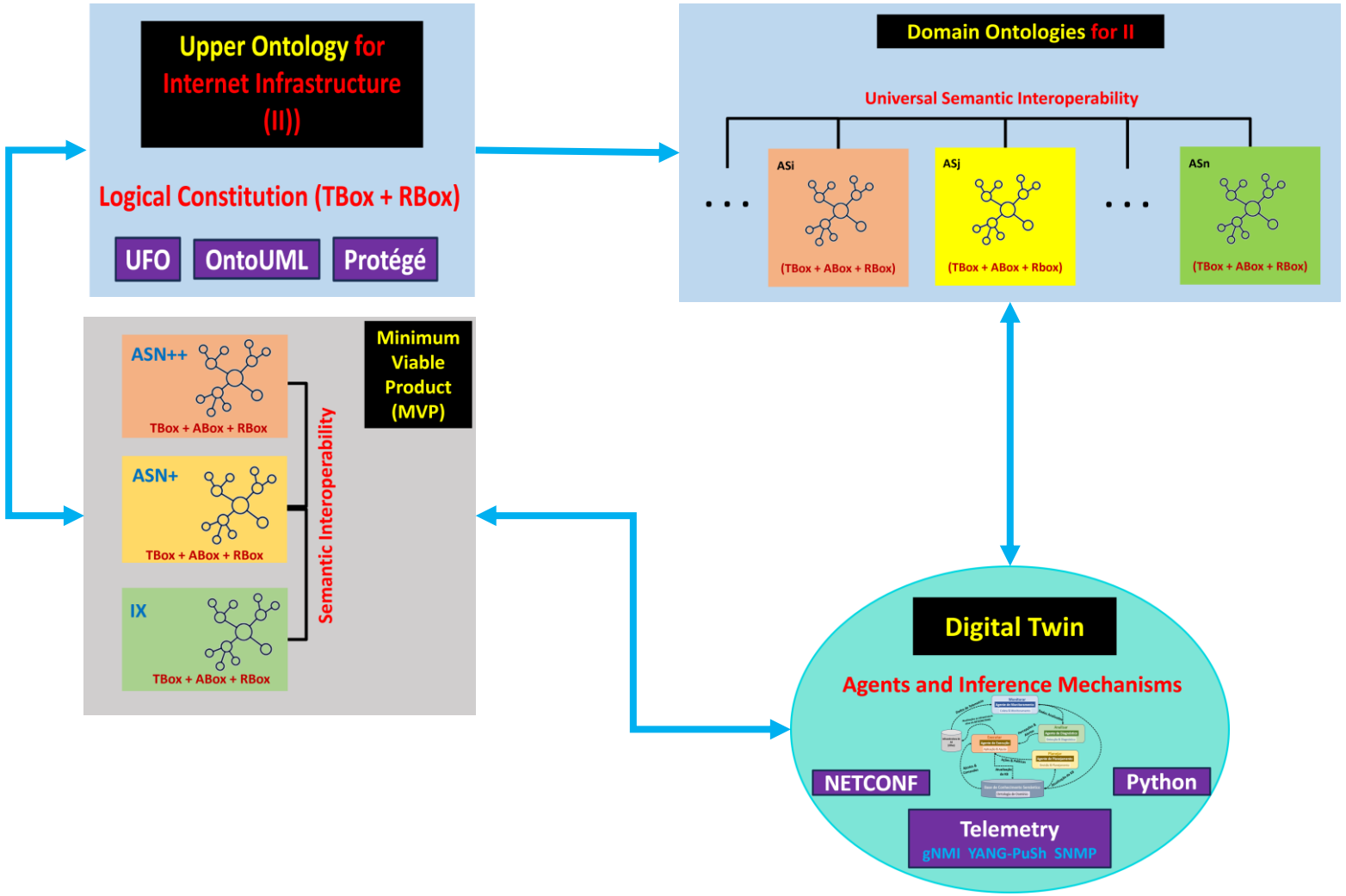
The Project


Domain Ontologies for II
Universal Semantic Interoperability
ASi
(TBox + ABox + RBox)
ASj
(TBox + ABox + Rbox)
ASn
(TBox + ABox + RBox)
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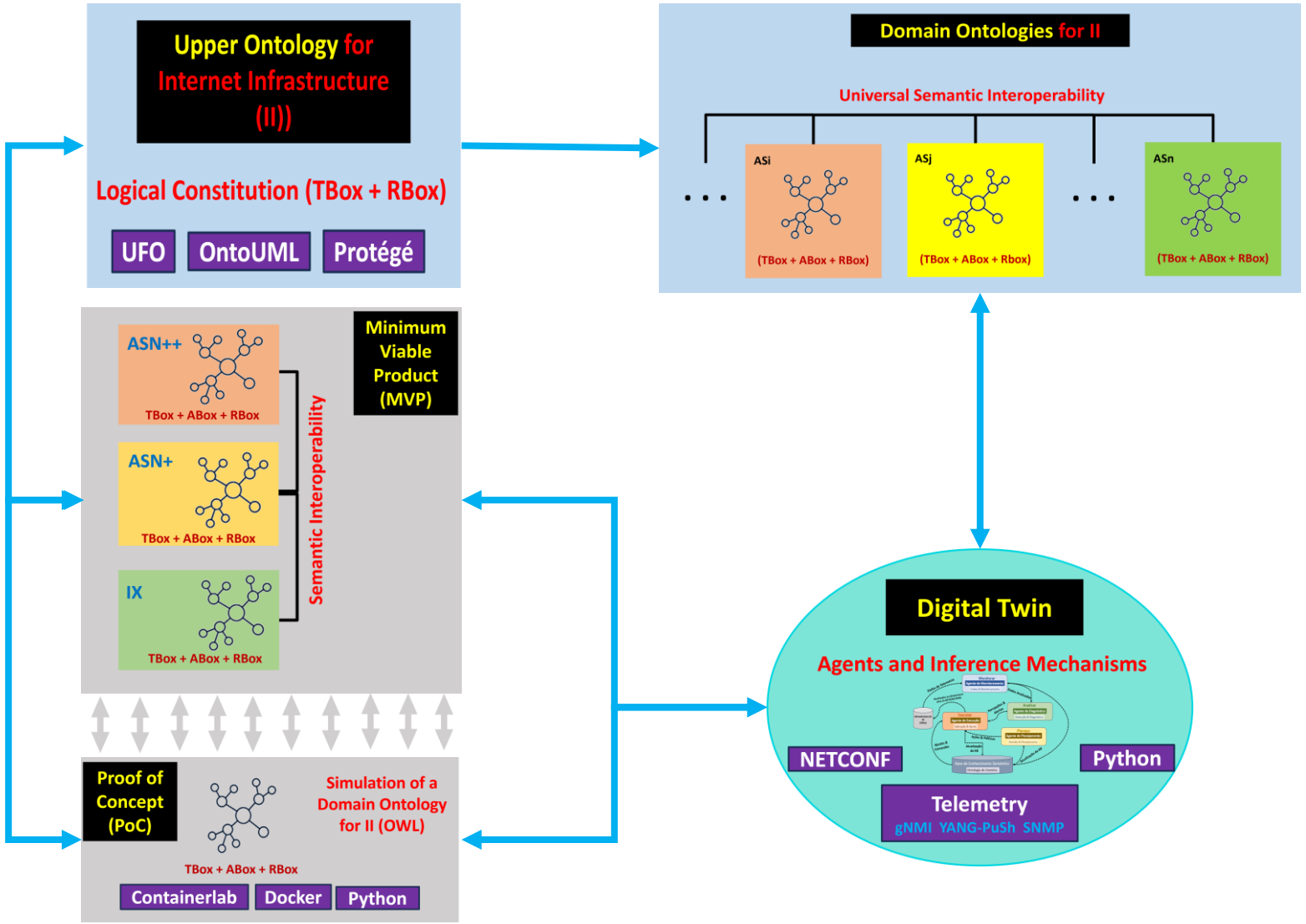
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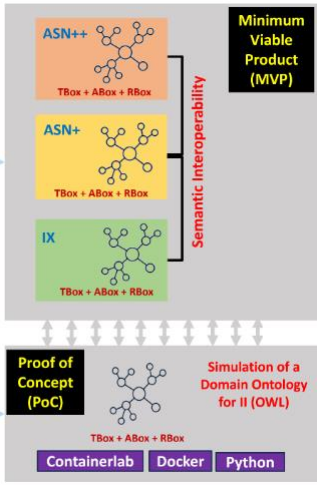

Upper Ontology for Internet Infrastructure (II))
Logical Constitution (TBox + RBox)
UFO
OntoUML
Protégé
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Minimum Viable Product (MVP)
ASN++
TBox + ABox + RBox
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IX
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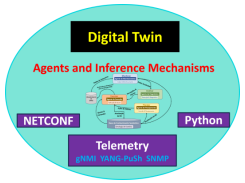
The Project

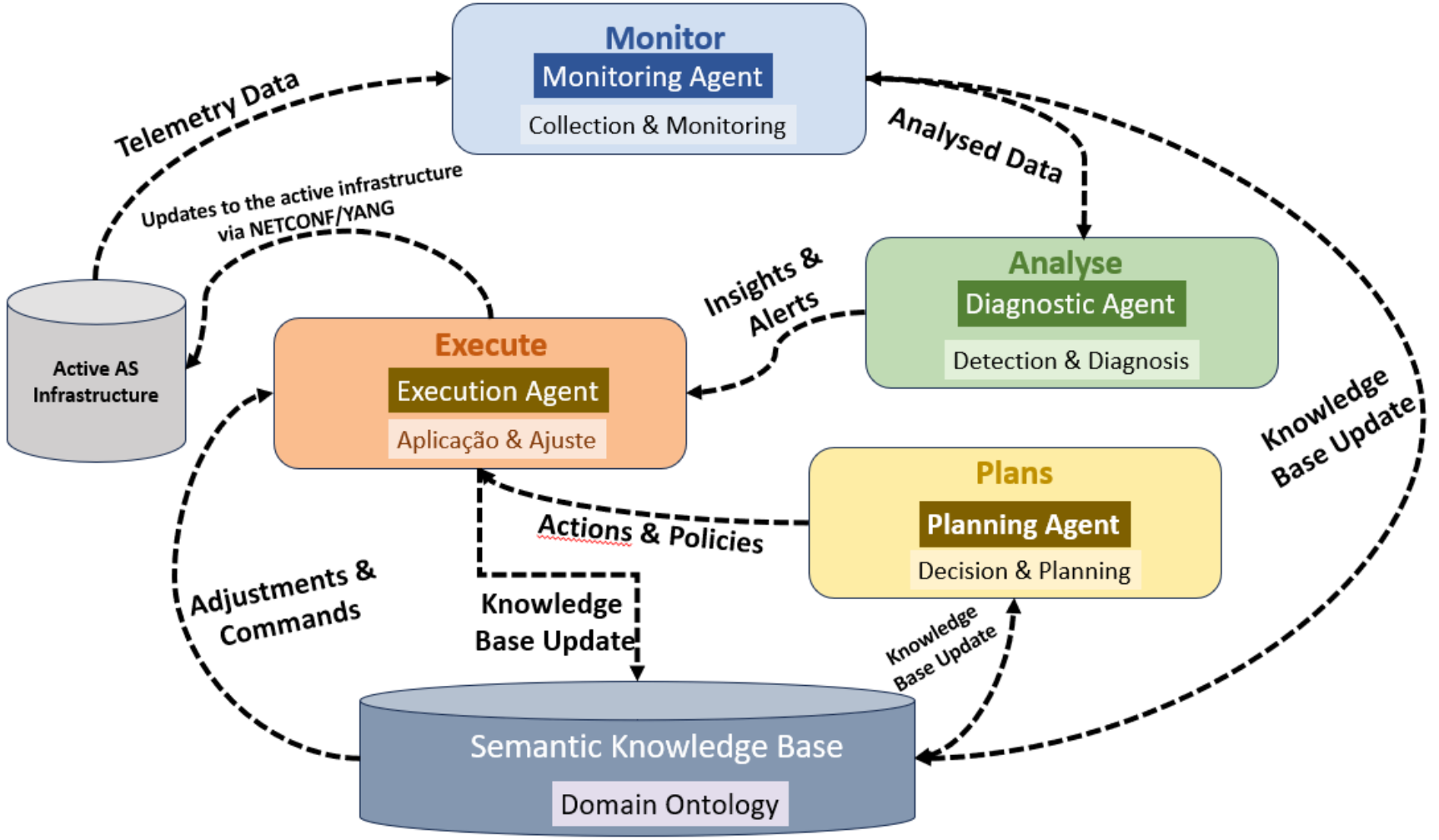

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Minimum Viable Product (MVP)
Proof of Concept (PoC)
Simulation of a Domain Ontology for II (OWL)
TBox + ABox + RBox
Containerlab
Docker
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Python
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gNMI YANG-PuSh SNMP



Classification of MVP participants, by level of complexity

High Complexity (ASN++)	Medium Complexity (ASN+)	Special Domain (IX)
Meet highly specialised and critical demands	Meet intermediate demands, with some level of specialisation	Meet a highly complex environment that centralises the interests of a large number of ASes of varying complexity
Robust infrastructure and advanced technology	Adequate infrastructure, but less sophisticated	Very complex infrastructure with highly diverse Internet traffic
Highly qualified multidisciplinary teams	Qualified technical teams, but less diverse	Highly qualified and diverse multidisciplinary teams
Services with high availability, redundancy, and security	Reliable services, but with less redundancy	Services with high availability and differentiated within the same environment
Integration with complex, large-scale systems	Moderate customisation capacity	Little or no integration capacity
Serve large corporations, public bodies, or strategic sectors	Serve medium-sized companies	Serve small, medium, large, and very large institutions with diverse service/-traffic demands
Bring the complexity of global routes, transit policies, and complex peering agreements	Bring the edge perspective, where regulations such as LGPD/SLA and local access policies apply	The neutral traffic exchange point, where the semantics of relators (multilateral BGP sessions, exchange agreements) are tested to the limit





MAPE-K

Jeffrey O. Kephart and David M. Chess. “The Vision of Autonomic Computing”. In: Computer. Vol. 36. 1. IEEE,2003, pp. 41–50. doi: 10.1109/MCOM.2003.1160055

DT Reference Architecture

Ontological Layer	**Tbox: define core classes (NetworkNode, Connection, Channel) and constraints**
Identity Layer	**Assigns stable IRIs to instances, separating identity from state**
Ingestion Layer	**Collects telemetry (YANG-Push, gNMI, SNMP, syslog) and normalises data into the Node–Connection–Channel triad**
Persistence Layer	**ABox + Rbox: stores facts in RDF/OWL and maintains temporal history via observations**
Validation Layer	**Applies structural constraints and logical consistency checks with DL and DeL reasoners**
Reasoning Layer	**Infers states and propagates failures**
Exposure Layer	**Provides SPARQL queries, dashboards, alerts, and integration with NOC/orchestration systems**

Meaning-Driven Regulatory Compliance

*Jurisdiction is no longer a property of hardware but a **normative relationship** (a relator) linking a norm, the parties and the network resources.*

SCENARIO



Use Case (I)

Rule (LGPD + GDPR): health data may only traverse adequately protected jurisdictions.

T — TBox
General rules

Health-data flows only traverse adequately protected jurisdictions.

R — RBox
How relations propagate

If a node is subject to the LGPD, so are the channels and connections passing through it.

A — ABox
Facts of the moment

Flow F carries health data · F passes through node N3 · N3 is under jurisdiction J.

The Digital Twin in Action (Use Case I)

Telemetry
gNMI · YANG-Push · SNMP

→

Semantic Elevation
raw data → meaning

→

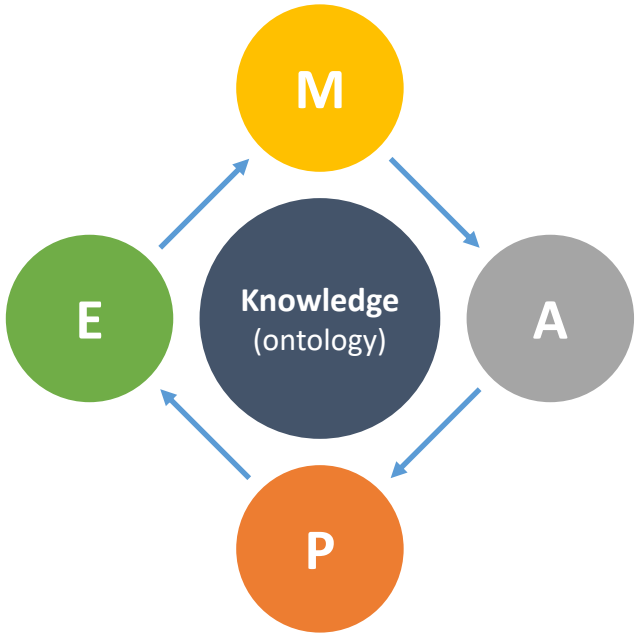
ABox
facts with meaning

→

Reasoner
checks the rules

Use Case (I)

MAPE-K LOOP AFTER A BGP FAILURE



- **M** — **Monitor:** Flow F has changed route.
- **A** — **Analyse:** F now traverses a non-compliant jurisdiction.
- **P** — **Plan:** A compliant alternative via AS-D is found.
- **E** — **Execute:** Applied via NETCONF, only after logical validation.

Takeaway: the network becomes jurisdiction-aware; compliance is checked by logical inference, not manual audits.

Responding to a DDoS Attack

*Conventional defences react to the **symptom** (suspicious IPs). DT-II reacts to the **meaning**: which services depend on what, and which agreements apply.*

Use Case (II)

SCENARIO

External botnet
attack source

→

Link_01
1 → 100 Gbps · saturated

→

DNS service
critical

→

Users
everything else depends on DNS

THE MAPE-K RESPONSE

M — Monitor
Steps 1–2

Telemetry becomes ABox facts: the flow is anomalous, Link_01 is saturated.

A — Analyse
Steps 3–4

The reasoner infers that DNS is at risk and identifies the source AS and its agreement.

P — Plan
Steps 5–7

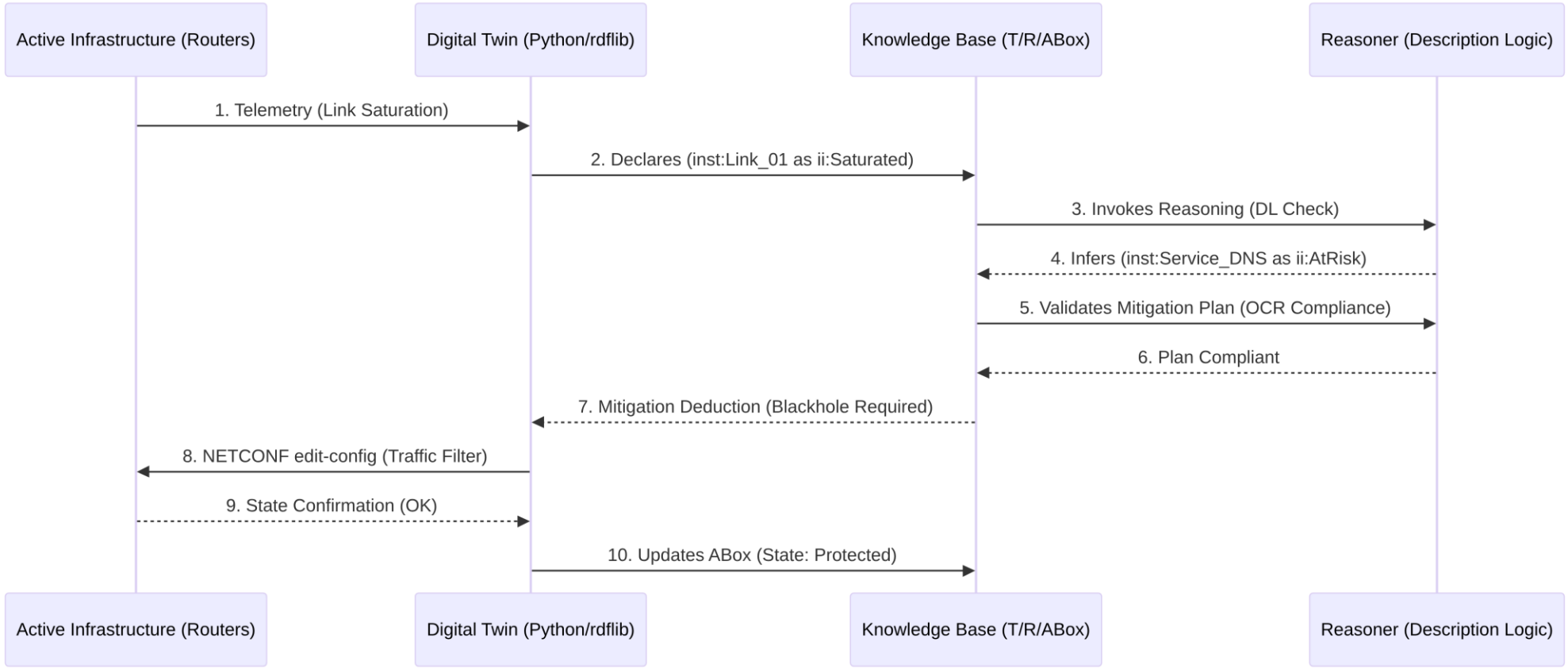
TBox rule: critical service at risk + external source → mitigate at the edge. The plan is validated first.

E — Execute
Steps 8–10

Filter applied via NETCONF, confirmed, and the ABox updated to “protected”.

Sequence Flow of the Mitigation

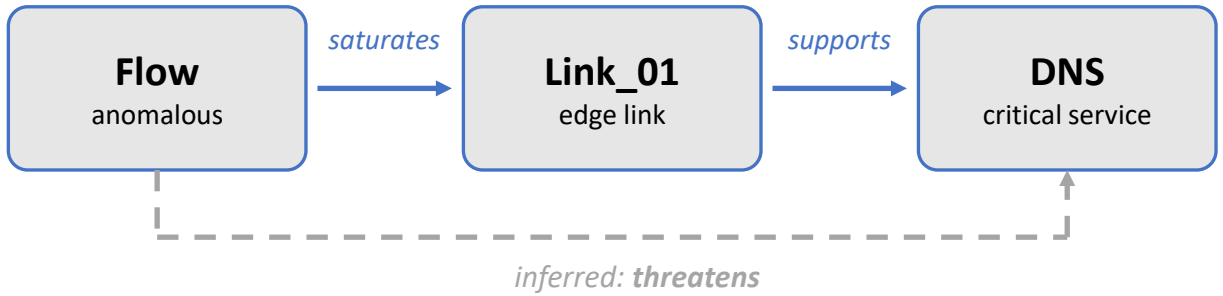
Use Case (II)



Monitor Steps 1–2 | **Analyse** Steps 3–4 | **Plan** Steps 5–7 | **Execute** Steps 8–10

Why a Semantic Defence?

HOW THE REASONER INFERS THE RISK



saturates ∘ supports ⊑ threatens

If a flow saturates a link and the link supports a service, the flow threatens the service. Nobody declares that DNS is at risk: the reasoner concludes it.

ADVANTAGES OVER IP BLOCKING

1 Context awareness

If the source AS has a cooperation agreement, ask it to filter at source instead of just blocking.

2 Mereological resilience

The model knows the parts of the network and their roles: sacrifice non-vital channels, keep DNS.

3 Traceability

Every action is explained by the axioms that justified it.

4 Compliance

No configuration is applied before it is validated against the compliance regime (Use Case I).

Use Case (II)

Takeaway: the defence knows what is at stake and why it acts, not just which IPs to block.

Contributions

- an ontology-driven reference model for II centred on Node–Connection–Channel, grounded in **UFO** and developed in **OntoUML**;
- an operationalisation in **DL/OWL** supporting automated reasoning over **TBox**, **ABox**, and **RBox**;
- a synchronisation approach aligning the KG with model-driven telemetry (e.g., **YANG-Push**) for near real-timestate updates;
- a governance mechanism based on an **Ontological Compliance Regime** and **SLA** constraints, enabling auditable and explainable compliance checks;
- a multi-agent Digital Twin architecture following the **MAPE-K** cycle for policy-driven actions;
- Second edition of the book **Knowledge Bases**, including an English language version. It incorporates knowledge gained through the research project.



Project Documentation

Project Digital Twin for Internet Infrastructure

Main: https://github.com/LJuliaoBraga/dtii

English: https://github.com/LJuliaoBraga/dtii/blob/main/docs/en-us/2026-ProjectDT-II-EN.pdf
Portuguese: https://github.com/LJuliaoBraga/dtii/blob/main/docs/pt-br/2026-ProjectDT-II-PT.pdf

ANMS 2006: https://github.com/LJuliaoBraga/dtii/blob/main/docs/events/anms%202026/AporesentacaoANMS-V3.pdf

Thank you